

Chapter 5

Random Vector : $X = [X_1, X_2, \dots, X_n]^T$ where X_i is a RV.

↓
random vector
with $n=1$.

Sample Value Vector : Sample values of the components of a random vector. $X = [x_1, x_2, \dots, x_n]^T$ where x_i is a sample value of the RV, X_i .

$$F_{X_1, \dots, X_n}(x_1, \dots, x_n)$$

CDF of a random vector

$$P_{X_1, \dots, X_n}(x_1, \dots, x_n)$$

PMF " " " "

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n)$$

PDF " " " "

Joint Probability Model of Two Random Vectors

$$F_{X,Y}(x,y) = F_{X_1, \dots, X_n, Y_1, \dots, Y_m}(x_1, \dots, x_n, y_1, \dots, y_m)$$

$$P_{X,Y}(x,y) = P_{X_1, \dots, X_n, Y_1, \dots, Y_m}(x_1, \dots, x_n, y_1, \dots, y_m)$$

$$f_{X,Y}(x,y) = f_{X_1, \dots, X_n, Y_1, \dots, Y_m}(x_1, \dots, x_n, y_1, \dots, y_m)$$

Ex: 5.6

$$f_X(x) = \begin{cases} 6e^{-a^T x} & , x > 0 \\ 0 & , \text{o/t} \end{cases}$$

where $a^T = [1, 2, 3]$, CDF ?

Ans:

$$a^T x = x [1, 2, 3]^T$$

$$= [1, 2, 3] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$= x_1 + 2x_2 + 3x_3$$

$$f_X(x) = 6e^{-(x_1 + 2x_2 + 3x_3)} , x_1, x_2, x_3 > 0$$

$$F_X(x) = \int_0^{x_1} \int_0^{x_2} \int_0^{x_3} 6e^{-(u_1 + 2u_2 + 3u_3)} du_3 du_2 du_1$$

$$= 6 \left(\int_0^{x_1} e^{-u_1} du_1 \right) \left(\int_0^{x_2} e^{-2u_2} du_2 \right) \left(\int_0^{x_3} e^{-3u_3} du_3 \right)$$

$$= 6 (1 - e^{-x_1}) \times \frac{1}{2} (1 - e^{-2x_2}) \times \frac{1}{3} (1 - e^{-3x_3})$$

$$= (1 - e^{-x_1}) (1 - e^{-2x_2}) (1 - e^{-3x_3})$$

Independent Random Vectors

$$P_{X,Y}(x,y) = P_X(x) P_Y(y)$$

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

Ex 5.7

$$f_{Y_1, \dots, Y_4}(y_1, \dots, y_4) = \begin{cases} 4 & , 0 \leq y_1 \leq y_2 \leq 1, 0 \leq y_3 \leq y_4 \leq 1 \\ 0 & , \text{otherwise} \end{cases}$$

$$\text{Let } V = [Y_1, Y_4]'$$

$$W = [Y_2, Y_3]'$$

Are V and W independent
Random Vectors?

Ans: Let $V = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$

$$0 \leq V_1 \leq W_1 \leq 1$$

$$W = \begin{bmatrix} W_1 \\ W_2 \end{bmatrix}$$

$$0 \leq W_2 \leq V_2 \leq 1$$

$$f_{V,W}(v,w) = \begin{cases} 4 & , 0 \leq v_1 \leq w_1 \leq 1, 0 \leq w_2 \leq v_2 \leq 1 \\ 0 & , \text{otherwise} \end{cases}$$

$$f_V(v) = \int_{v_1}^1 \int_0^{v_2} 4 \, dw_2 \, dw_1$$

$$f_V(v) = 4(1-v_1)v_2 \quad , 0 \leq v_1, v_2 \leq 1$$

$$f_W(w) = \int_0^{w_1} \int_{w_2}^1 4 \, dv_2 dv_1$$

$$f_W(w) = 4w_1(1-w_2), \quad 0 \leq w_1, w_2 \leq 1$$

For independence,

$$f_{V,W}(v,w) = f_V(v) \times f_W(w)$$

$$\begin{aligned} \text{Thus, } f_V(v) \times f_W(w) &= 4(1-v_1)v_2 \times 4w_1(1-w_2) \\ &= 16(1-v_1)(1-w_2)v_2w_1 \neq f_{V,W}(v,w) \end{aligned}$$

Thus V & W are not independent. (Ans)

$$\underline{E[(X-\mu)(X-\mu)^T] = T(X-\mu)(X-\mu)^T}$$

Expected value vector: $E[(X-\mu)(X-\mu)^T] = T(X-\mu)(X-\mu)^T$

$$\text{If } X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \text{ then } E[X] = \mu_X = \begin{bmatrix} E[x_1] \\ \vdots \\ E[x_n] \end{bmatrix}$$

$$\begin{bmatrix} (x_1 - \mu_1)(x_1 - \mu_1) & (x_1 - \mu_1)(x_2 - \mu_2) \\ (x_2 - \mu_2)(x_1 - \mu_1) & (x_2 - \mu_2)(x_2 - \mu_2) \\ \vdots & \vdots \end{bmatrix}$$

Take expectation

$$\begin{bmatrix} E[(x_1 - \mu_1)(x_1 - \mu_1)] & E[(x_1 - \mu_1)(x_2 - \mu_2)] \\ E[(x_2 - \mu_2)(x_1 - \mu_1)] & E[(x_2 - \mu_2)(x_2 - \mu_2)] \\ \vdots & \vdots \end{bmatrix}$$

Covariance of a Random Vectors

for ordinary RV's, $\text{Cov}(X_i, X_j) = E[(X_i - E[X_i])(X_j - E[X_j])]$

for a random vector, we want the covariance between every possible pair of components.

$$\text{Cov}[X] = E[(X - \mu_X)(X - \mu_X)^T]$$

For $n=2$, $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ $\mu_X = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$

$$X - \mu_X = \begin{bmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \end{bmatrix}$$

$$(X - \mu_X)^T = [X_1 - \mu_1 \quad X_2 - \mu_2]$$

$$(X - \mu_X)(X - \mu_X)^T = \begin{bmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \end{bmatrix} [X_1 - \mu_1 \quad X_2 - \mu_2]$$

$$= \begin{bmatrix} (X_1 - \mu_1)^2 & (X_1 - \mu_1)(X_2 - \mu_2) \\ (X_2 - \mu_2)(X_1 - \mu_1) & (X_2 - \mu_2)^2 \end{bmatrix}$$

Take expectations :

$$E[(X - \mu_X)(X - \mu_X)^T] = \begin{bmatrix} E[(X_1 - \mu_1)^2] & E[(X_1 - \mu_1)(X_2 - \mu_2)] \\ E[(X_2 - \mu_2)(X_1 - \mu_1)] & E[(X_2 - \mu_2)^2] \end{bmatrix}$$

$$\text{Cov}[X] = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) \end{bmatrix}$$

$\hookrightarrow$ since $E[(X_i - \mu_i)^2] = \text{Var}(X_i)$
 and $E[(X_i - \mu_i)(X_j - \mu_j)] = \text{Cov}(X_i, X_j)$

General form

$$\text{Cov}[X] = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \dots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \dots & \text{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \text{Cov}(X_n, X_2) & \dots & \text{Var}(X_n) \end{bmatrix}$$

$$\text{Cov}(X_i, X_i) = \text{Var}(X_i)$$

$$\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$$

$$\begin{bmatrix} X_1 & X_2 & \dots & X_n \\ X_1 & X_2 & \dots & X_n \\ \vdots & \vdots & \ddots & \vdots \\ X_1 & X_2 & \dots & X_n \end{bmatrix} = \begin{bmatrix} X_1 & X_2 & \dots & X_n \\ X_1 & X_2 & \dots & X_n \\ \vdots & \vdots & \ddots & \vdots \\ X_1 & X_2 & \dots & X_n \end{bmatrix} = X^T X$$

$$E[X X^T] = \begin{bmatrix} E[X_1^2] & E[X_1 X_2] & \dots & E[X_1 X_n] \\ E[X_2 X_1] & E[X_2^2] & \dots & E[X_2 X_n] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_n X_1] & E[X_n X_2] & \dots & E[X_n^2] \end{bmatrix} = \text{Cov}[X]$$

$\hookrightarrow$ Covariance matrix

General Form - II

$$\text{Cov}[x] = E[(x - \mu_x)(x - \mu_x)^T]$$

$$= E[(x - \mu_x)(x^T - \mu_x^T)]$$

$$= E[(xx^T - x\mu_x^T - \mu_x x^T + \mu_x \mu_x^T)]$$

$$= E[xx^T] - E[x\mu_x^T] - E[\mu_x x^T] + E[\mu_x \mu_x^T]$$

$$= E[xx^T] - \mu_x^T E[x] - \mu_x E[x^T] + \mu_x \mu_x^T$$

$$= E[xx^T] - \mu_x^T \mu_x - \mu_x \mu_x^T + \mu_x \mu_x^T$$

$$\text{Cov}[x] = E[xx^T] - \mu_x \mu_x^T$$

What is xx^T ?

Suppose, $n=2$ $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$xx^T = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} x_1^2 & x_1 x_2 \\ x_2 x_1 & x_2^2 \end{bmatrix}$$

$$E[xx^T] = \begin{bmatrix} E[x_1^2] & E[x_1 x_2] \\ E[x_2 x_1] & E[x_2^2] \end{bmatrix} = R_x$$

↳ Correlation Matrix

Thus $\text{Cov}(X) = R_X - \mu_X \mu_X^T$

Ex 8.5 (pg 19) Self-Proc.

Cross-Correlation of Two Random Vectors

Let $X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ and $Y = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}$

$$R_{xy} = E[XY^T]$$

cross correlation matrix

$$R_{xy} = \begin{bmatrix} E[x_1 y_1] & E[x_1 y_2] \\ E[x_2 y_1] & E[x_2 y_2] \end{bmatrix}, \text{ for } n=2, m=2$$

Thus, $R_{xy}(i, j) = E[x_i y_j]$

Cross - Covariance

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)^T] \quad \text{where } X \text{ and } Y \text{ are random vectors}$$

$$\text{Cov}(X, Y) = \begin{bmatrix} \text{Cov}(X_1, Y_1) & \text{Cov}(X_2, Y_2) \\ \text{Cov}(X_1, Y_2) & \text{Cov}(X_2, Y_2) \end{bmatrix}$$

$$\text{Thus } \text{Cov}(X, Y)_{ij} = \text{Cov}(X_i, Y_j)$$

Or:

$$\text{Cov}(X, Y) = R_{XY} - \mu_X \mu_Y^T$$

Affine Transformation

for a random vector X with $\dim(n \times 1)$,

$$Y = AX + b$$

matrix with $\dim(m \times n)$ vector with $\dim(m \times 1)$

New vector made from linear combinations of the components of X , plus constants (affine transformation)

$$\mu_y = A\mu_x + b$$

$$\text{Cov}(Y) = A \times \text{Cov}(X) \times A^T$$

$$R_y = AR_x A^T + (A\mu_x)b^T + b(A\mu_x)^T + bb^T$$

better derive from :

$$C_y = R_y - \mu_y \mu_y^T$$

$$\Rightarrow R_y = C_y + \mu_y \mu_y^T$$

$$\Rightarrow R_y = AC_x A^T + (A\mu_x + b)(A\mu_x + b)^T$$

Ex 5.9

Given $\mu_x = \begin{bmatrix} 1/2 \\ 2/3 \end{bmatrix}$ $C_x = \begin{bmatrix} 1/18 & 1/36 \\ 1/36 & 1/18 \end{bmatrix}$

$R_x = \begin{bmatrix} 1/6 & 1/4 \\ 1/4 & 1/2 \end{bmatrix}$

$Y = AX + b$ where $A = \begin{bmatrix} 1 & 0 \\ 6 & 3 \\ 3 & 6 \end{bmatrix}$

$b = \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix}$

Find μ_y, C_y, R_y .

Ans: $\mu_y = A\mu_x + b$

$= \begin{bmatrix} 1 & 0 \\ 6 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1/2 \\ 2/3 \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix}$

$= \begin{bmatrix} 1/3 \\ 2 \\ 3 \end{bmatrix}$

$C_y = AC_x A^T$

$= \begin{bmatrix} 1 & 0 \\ 6 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1/18 & 1/36 \\ 1/36 & 1/18 \end{bmatrix} \begin{bmatrix} 1 & 6 & 3 \\ 0 & 3 & 6 \end{bmatrix}$

$$= \begin{bmatrix} \overset{\text{var}(Y_1)}{1/8} & \overset{\text{Cov}(Y_1, Y_2)}{5/12} & 1/3 \\ 5/12 & 7/2 & 13/4 \\ 1/3 & 13/4 & 7/2 \end{bmatrix}$$

$$C_Y = R_Y - \mu_Y \mu_Y^T$$

$$R_Y = C_Y + \mu_Y \mu_Y^T$$

$$\mu_Y \mu_Y^T = \begin{bmatrix} 1/3 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1/3 & 2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1/9 & 2/3 & 1 \\ 2/3 & 4 & 6 \\ 1 & 6 & 9 \end{bmatrix}$$

$$R_Y = \begin{bmatrix} 1/8 & 5/12 & 1/3 \\ 5/12 & 7/2 & 13/4 \\ 1/3 & 13/4 & 7/2 \end{bmatrix} + \begin{bmatrix} 1/9 & 2/3 & 1 \\ 2/3 & 4 & 6 \\ 1 & 6 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 1/6 & 13/12 & 4/3 \\ 13/12 & 15/2 & 37/4 \\ 4/3 & 37/4 & 25/2 \end{bmatrix}$$

PDF of a Transformed Random Vector

$$\text{If } Y = AX + b$$

and A is invertible, then

$$f_Y(y) = \frac{1}{|\det(A)|} f_X(A^{-1}(y-b))$$

Cross Correlation between X and Y

$$R_{xy} = E[XY^T]$$

$$= E[X(AX+b)^T]$$

$$= E[X(X^T A^T + b^T)]$$

$$= E[XX^T A^T + Xb^T]$$

$$= E[XX^T] \cdot A^T + E[X] b^T$$

A^T and b^T are constants

$$\boxed{R_{xy} = R_x A^T + \mu_x b^T}$$

Cross Covariance

$$C_{xy} = E[(x - \mu_x)(y - \mu_y)^T]$$

$$= E[(x - \mu_x)(Ax + b - (A\mu_x + b))^T]$$

$$= E[(x - \mu_x)A(x - \mu_x)^T]$$

$$= E[(x - \mu_x)(x - \mu_x)^T A^T]$$

$$= E[(x - \mu_x)(x - \mu_x)^T] A^T$$

$$\boxed{C_{xy} = C_x A^T}$$

$$\begin{bmatrix} 1 \times \\ n \times \end{bmatrix} = X$$

Correlation Coefficient

$$\rho(u, v) = \frac{\text{Cov}(u, v)}{\sqrt{\text{Var}(u)\text{Var}(v)}} \quad \text{and } -1 \leq \rho \leq 1$$

↑
scalar ρ between
two RVs, u and v

For random vectors X and Y

$$\rho_{X_2, Y_1} = \frac{\text{Cov}(X_2, Y_1)}{\sqrt{\text{Var}(X_2) \text{Var}(Y_1)}}$$

← value from $C_{xy}(2,1)$

value from $C_X(2,2)$ value from $C_Y(1,1)$

Gaussian Random Vector

Multivariable version of the normal distribution.

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

If X is Gaussian, $X \sim \mathcal{N}(\mu_X, C_X)$ $\mu_X = E[X]$

$$C_X = \text{Cov}[X]$$

*** If a random vector is Gaussian and

C_X is a diagonal matrix then X 's components are independent.

Affine transformation of a Gaussian Random Vector also results in a Gaussian Random Vector.

$$f_X(u) = \frac{1}{(2\pi)^{n/2} \times \sqrt{\det(C_X)}} \exp\left(-\frac{1}{2}(u-\mu_X)^T C_X^{-1} (u-\mu_X)\right)$$

↑
ensures total
prob. integrates
to 1

Ex 8.10

Given $\mu_X = \begin{bmatrix} 50 \\ 62 \\ 68 \end{bmatrix}$ $C_X = \begin{bmatrix} 16 & 12.8 & 11.2 \\ 12.8 & 16 & 12.8 \\ 11.2 & 12.8 & 16 \end{bmatrix}$

$$\begin{aligned} \text{Var}(X_1) &= \text{Var}(X_2) = \text{Var}(X_3) \\ &= 16 \\ \sigma_1 &= \sigma_2 = \sigma_3 = 4 \end{aligned}$$

a) $W = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$, $f_W(w) = ?$

$$\mu_W = \begin{bmatrix} 50 \\ 62 \end{bmatrix}$$

$$C_W = \begin{bmatrix} 16 & 12.8 \\ 12.8 & 16 \end{bmatrix}$$

$$\det(C_W) = 16(16) - 12.8(12.8)$$

$$= 92.16$$

$$\sqrt{\det(C_W)} = \sqrt{92.16} = 9.6$$

$$C_W^{-1} = \frac{1}{\det(C_W)} \begin{bmatrix} 16 & -12.8 \\ -12.8 & 16 \end{bmatrix} = \frac{1}{92.16} \begin{bmatrix} 16 & -12.8 \\ -12.8 & 16 \end{bmatrix}$$

$$= \frac{1}{92.16} \begin{bmatrix} 16 & -12.8 \\ -12.8 & 16 \end{bmatrix}$$

$$w - \mu_W = \begin{bmatrix} x_1 - 50 \\ x_2 - 62 \end{bmatrix}$$

$$f_W(w) = \frac{1}{(2\pi)^{2/2} \times \det(C_W)^{1/2}} \exp\left(-\frac{1}{2} (w - \mu_W)^T C_W^{-1} (w - \mu_W)\right)$$

$$= \frac{1}{2\pi \times 9.6} \times \exp\left(-\frac{1}{2} \begin{bmatrix} x_1 - 50 \\ x_2 - 62 \end{bmatrix} \times \frac{1}{92.16} \begin{bmatrix} 16 & -12.8 \\ -12.8 & 16 \end{bmatrix} \times \begin{bmatrix} x_1 - 50 \\ x_2 - 62 \end{bmatrix}\right)$$

$$\text{let, } a = x_1 - 50 \\ b = x_2 - 62$$

$$f_W(w) = \frac{1}{60.3} \times \exp\left(-\frac{1}{2} \begin{bmatrix} a \\ b \end{bmatrix} \times \frac{1}{92.16} \begin{bmatrix} 16 & -12.8 \\ -12.8 & 16 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}\right)$$

$$= \frac{1}{60.3} \exp\left(-\frac{a^2 - 16ab + b^2}{11.52}\right)$$

$$= \frac{1}{60.3} \exp\left(-\frac{(x_1 - 50)^2 - 16(x_1 - 50)(x_2 - 62) + (x_2 - 62)^2}{11.52}\right)$$

Shortcut :

Let $\rho_{x_1, x_2} = \rho$

$$f_{x_1, x_2}(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\right)$$

$$\times \left(\frac{(x_1 - \mu_1)^2}{\sigma_1^2} - \frac{2\rho(x_1 - \mu_1)(x_2 - \mu_2)}{\sigma_1\sigma_2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} \right)$$

Ex : 8.11

$$Y_i = \left(\frac{5}{9}\right)(X_i - 32)$$

Temp in
°C

Temp in
°F

a)

$$\Rightarrow Y_i = \frac{5}{9} X_i - \frac{160}{9}$$

$$Y = AX + b \quad \text{with } A = \begin{bmatrix} 5/9 & 0 & 0 \\ 0 & 5/9 & 0 \\ 0 & 0 & 5/9 \end{bmatrix} = 5/9 I$$

$$\text{and } b = -\frac{160}{9} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \neq$$

$$\mu_y = A \mu_x + b$$

$$= \frac{5}{9} I \times \begin{bmatrix} 50 \\ 62 \\ 58 \end{bmatrix} - \frac{160}{9} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 10 \\ 50/3 \\ 130/9 \end{bmatrix} (Am) = (18.111) \times 10^3$$

b) $C_y = ?$

$$C_y = A C_x A^T$$

$$= \frac{5}{9} I \times C_x \times (\frac{5}{9} I)^T$$

$$= \frac{5}{9} I \times C_x \times \frac{5}{9} I$$

$$= \frac{25}{81} \times C_x$$

$$= \frac{25}{81} \times \begin{bmatrix} \frac{400}{81} & \frac{320}{81} & \frac{280}{81} \\ \frac{320}{81} & \frac{400}{81} & \frac{320}{81} \\ \frac{280}{81} & \frac{320}{81} & \frac{400}{81} \end{bmatrix} (Am)$$

c) $f_Y(y) = ?$

$$f_Y(y) = \frac{1}{(2\pi)^{n/2} \det(C_Y)^{1/2}} \exp\left(-\frac{1}{2}(y - \mu_Y)^T C_Y^{-1} (y - \mu_Y)\right)$$

$$= \frac{1}{(2\pi)^{3/2} \det(C_Y)^{1/2}}$$

$$n=3$$

$$\det(kA) = k^n \det(A)$$

$$\det(C_Y) = \det(25/81 \times C_X)$$

$$= \left(\frac{25}{81}\right)^3 \det(C_X)$$

$$\sqrt{\det(C_Y)} = \sqrt{\left(\frac{25}{81}\right)^3 \times \det(C_X)}$$

$$= \sqrt{\left(\frac{25}{81}\right)^3 \times 516.696}$$

$$= \cancel{22.717} \times 3.895$$

$$(2\pi)^{3/2} \times \frac{3.895}{\cancel{22.717}} = \cancel{61.35} \times 61.35$$

$$f_Y(y) = \frac{1}{61.35} \exp\left(-\frac{1}{2}(y - \mu_Y)^T (25/81 C_X)^{-1} (y - \mu_Y)\right)$$

$$f_Y(y) = \frac{1}{\sigma_Y} \exp\left(-\frac{1}{2\sigma_Y^2}(y-\mu_Y)^2\right) \quad (\text{Ans})$$

Jointly Gaussian

If two RV's are jointly Gaussian,

$$E[X] = \mu_X$$

$$E[Y] = \mu_Y$$

$$\Sigma = \sigma$$

$$\text{Var}(X) = \sigma_X^2$$

$$\text{Var}[Y] = \sigma_Y^2$$

corr. coeff : ρ_{XY}

X and Y are by themselves gaussian too.

$$X \sim N(\mu_X, \sigma_X^2)$$

$$Y \sim N(\mu_Y, \sigma_Y^2)$$

$X|Y=y$ is Gaussian too.

conditional mean,

$$\mu_{X|Y} = \mu_X + \rho_{XY} \frac{\sigma_X}{\sigma_Y} (y - \mu_Y)$$

conditional variance,

$$\sigma_{X|Y}^2 = \sigma_X^2 (1 - \rho_{XY}^2)$$

conditional variance does not depend on observed value of y

Generally, if x, y independent $\Rightarrow \rho_{xy} = 0$

However, if x and y are jointly Gaussian,

$$x, y \text{ independent} \Leftrightarrow \rho_{xy} = 0$$

Mahalanobis Distance

$$\Delta^2 = (x - \mu_x)^T C_x^{-1} (x - \mu_x)$$

$$\text{Therefore, } f_x(x) = \frac{1}{(2\pi)^{n/2} \sqrt{|C_x|}} \exp\left[-\frac{1}{2} (x - \mu_x)^T C_x^{-1} (x - \mu_x)\right]$$

$$\text{becomes, } f_x(x) = \frac{1}{(2\pi)^{n/2} \sqrt{|C_x|}} e^{-\Delta^2/2}$$